

2024



VIVOTEK Accelerates Launch of VORTEX, Its Cloud Surveillance Platform, by 12 Months with AWS IoT Core

VIVOTEK launches its next-generation video-surveillance-as-a-service (VSaaS) solution, VORTEX, on AWS IoT Core—saving time and R&D resources to focus on innovation and scalability.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

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Cost-effective innovation Overview

Enhancing AI-powered features

[VIVOTEK](#), a global IP security solution provider, opted for Amazon Web Services (AWS) as the cloud platform for its hybrid video surveillance-as-a-service solution, VORTEX. VIVOTEK adopted cloud technology early on and chose AWS for its reliability and scalability, with AWS already supporting VIVOTEK's existing software development and facilitating access to security cameras via mobile devices.

VORTEX used AWS services, including [AWS IoT Core](#), to successfully launch VORTEX in 12 months, offering a secure, scalable surveillance platform that efficiently processes and stores data from security cameras across the globe. As a result, the business reduced infrastructure management, dedicating 80 percent of its resources towards innovation, and effectively enhancing the solution's AI-powered features.



Opportunity | Ensuring Reliable Global Delivery of V Surveillance-as-a-Service Solutions



[VIVOTEK](#), established in 2000, provides security surveillance solutions worldwide. With headquarters in Taiwan, the company has offices and subsidiary businesses across the US, Europe, Japan, Latin America, and Asia. It specializes in manufacturing security equipment, particularly internet protocol (IP) cameras, which transmit video data across networks for central storage.

VIVOTEK revolutionized its architecture by using the cloud to develop video-surveillance-as-a-service (VSaaS) offerings. With VSaaS, customers store video data in a scalable cloud infrastructure, enabling multi-location management, centralized device control, remote monitoring, and a 24/7 security posture. VSaaS solutions also offer streamlined IT management by eliminating the need to maintain underlying hardware, while providing more intuitive browser-based interfaces. ShengFu Cheng, vice president of the Strategy Business Division at VIVOTEK, adds, "Deploying VSaaS solutions across multiple sites is now simpler, with one-click firmware over-the-air [OTA] upgrades as a routinely available feature."

In 2022 VIVOTEK redefined its solution with a hybrid model and developed VORTEX, a VSaaS platform integrating video management services (VMS), a high-quality artificial intelligence (AI) camera, and cloud backup features. To ensure seamless global delivery of VORTEX, VIVOTEK sought a reliable and scalable cloud infrastructure. Given VIVOTEK's successful engagement with Amazon Web Services (AWS) for its previous offering, VIVOCLOUD—a web-based solution facilitating access to IP cameras via smartphones and other mobile devices—AWS was a logical choice for this next-generation platform.

ShengFu says, "Since the launch of VIVOCLOUD in 2017, AWS has consistently proven its reliability, and the steadfast support from AWS teams has instilled us with confidence."

Solution | Saving 12 Months of Development Time with AWS IoT Core

When establishing the AWS infrastructure for VORTEX, VIVOTEK employed several of the same services utilized for [VIVOCLOUD](#). To ensure secure connections between IP cameras and AWS, the company implemented [AWS IoT Core](#), a service capable of connecting billions of Internet of Things (IoT) devices and routing trillions of messages to AWS services. Furthermore, the flexibility of AWS IoT Core meant VIVOTEK could adopt the MQTT (Message Queuing Telemetry Transport) protocol, an efficient and cost-effective method for transmitting data between devices and the cloud.



ShengFu says, “With AWS IoT Core, we can focus on our data streaming technology, analytics capabilities, and other core business areas. We don’t have to spend time managing devices.”

VIVOTEK runs VORTEX’s VSaaS applications in Kubernetes containers on [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances, with [Amazon Elastic Kubernetes Service](#) (Amazon EKS) automating container deployment, scaling, and management. To further simplify management, VORTEX uses [Amazon Relational Database Service](#) (Amazon RDS) for application databases, [Amazon DynamoDB](#) for unstructured data, and [Amazon Simple Storage Service](#) (Amazon S3) for video data storage and backups. [Amazon Simple Notification Service](#) (Amazon SNS) also sends push notifications to apps and customer devices.

VIVOTEK protects its AWS environments through [AWS WAF](#) and [Amazon GuardDuty](#), an intelligent threat detection service. Meanwhile, [Amazon Cognito](#) handles customer identity and access management at the infrastructure’s front end.

By using tried-and-tested AWS services, VIVOTEK significantly reduced development time—bringing VORTEX to market, including full-stack web and mobile apps, in just 12 months. “We saved a year in the development process by using AWS IoT Core for the launch of VORTEX, as it eliminated the need to design a solution for connecting cameras to the cloud,” ShengFu explains.

Outcome | Expanding AI-Powered Capabilities While Controlling Costs

In 2022, VIVOTEK introduced VORTEX on a subscription basis to its global customers. The launch came with advanced object recognition technology across people and vehicles for AI-powered video analytics, instant notifications, and false alarm minimization. Moreover, the VSaaS solution requires zero configuration, and customers can deploy the camera by scanning a QR code, enabling centralized remote monitoring and saving on-site setup time.

Following the launch, VORTEX is focused on enhancing the user experience to ensure an efficient operational environment and provide flexibility for business adaptation, all without requiring the replacement of existing cameras. This approach streamlines migration efforts and facilitates smoother transitions. VORTEX also provides features such as line-crossing detection, loitering detection, and path extraction. Moreover, it offers future capabilities aimed at expanding the surveillance ecosystem, enabling customers to leverage metadata and unlock actionable insights.

Looking ahead, the company aims to process image and cloud-based data more efficiently as it advances VORTEX’s AI-powered features. “We appreciate AWS’s commitment in aligning with our development goals and finding ways to improve performance without increasing costs,” says ShengFu.

ShengFu concludes, “We’re excited to continue enhancing VORTEX. More than 80 percent of our resources can go towards driving innovation because managing our infrastructure on AWS is so simple.”

About VIVOTEK

Established in Taiwan in 2000, VIVOTEK is known for its expertise in imaging and audio technology. The company provides a wide range of IP cameras, video management software, VSaaS solutions, and AI video analytics. VIVOTEK has been a long-time partner of AWS, and the company is proud to be an AWS IoT Core partner.



listed on the Taiwan Stock Exchange in 2011 and has since expanded its presence globally with offices in the US, Netherlands, India, Mexico, and Japan.

AWS Services Used

AWS IoT Core

AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

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Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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Amazon SNS

Amazon Simple Notification Service (Amazon SNS) sends notifications two ways, A2A and A2P. A2A provides high-throughput, push-based, many-to-many messaging between distributed systems, microservices, and event-driven serverless applications.

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AWS WAF

AWS WAF helps you protect against common web exploits and bots that can affect availability, compromise security, or consume excessive resources.

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